NEP Aligned

1ICPC317

AY 2024-25

# SDLC Laboratory

## Quality Laboratory Manual

**Experiment No. 01**

**To Understand the Phases in a Software Development Project**

**Course Instructor –**
**Dr. Tahseen A. Mulla and Prof. Amruta R. Patil**

# Experiment No. 01

**Title of Experiment:** To Understand the Phases in a Software Development Project (Software Development Life Cycle – SDLC)

**Aim of Experiment:** To study and understand the various phases involved in a software development project and their significance in delivering a quality software product

**System Requirements –**

- Operating System: Windows 10 or above
- RAM: Minimum 4 GB
- Processor: 2.33 GHz or higher

**Software/s Needed for Experiment**

- PDF viewer/ Office suite (MS Word/ LibreOffice)
- Web Browser (for reference)

**Experiment Objectives:**

- To understand the concept of Software Development Life Cycle (SDLC)
- To identify and describe each phase of a software development project
- To understand the role of documentation and validation in SDLC
- To analyze how SDLC phases contribute to successful project management

**Experiment Outcomes:**

After completing the experiment, students will be able to –

- Explain the need for SDLC in software engineering
- Identify different phases of SDLC and their deliverables
- Relate SDLC phases to real-world software projects
- Demonstrate basic understanding required for subsequent SDLC experiments

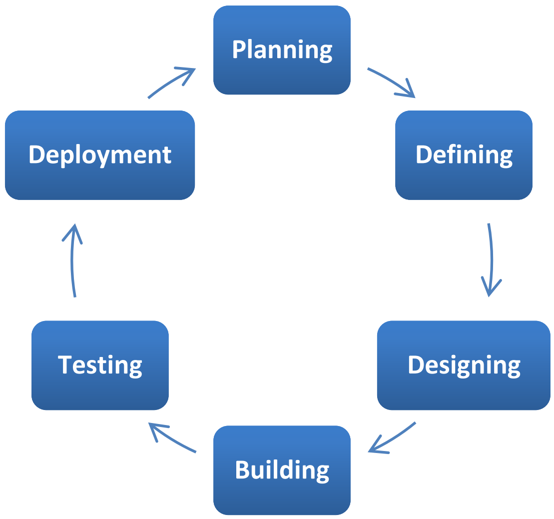
**Theory:**

The Software Development Life Cycle (SDLC) is a structured framework used by software organizations to design, develop, and test high-quality software. It defines the entire process of software production, from the initial idea to the final deployment and maintenance. The goal of the SDLC is to produce high-quality software that meets or exceeds customer expectations, reaches completion within time and cost estimates, and is efficient to maintain.

**Why is SDLC Important?**

Without a structured SDLC, software development becomes chaotic. Teams may miss requirements, blow budgets, or deliver buggy code that doesn't solve the user's problem. SDLC provides:

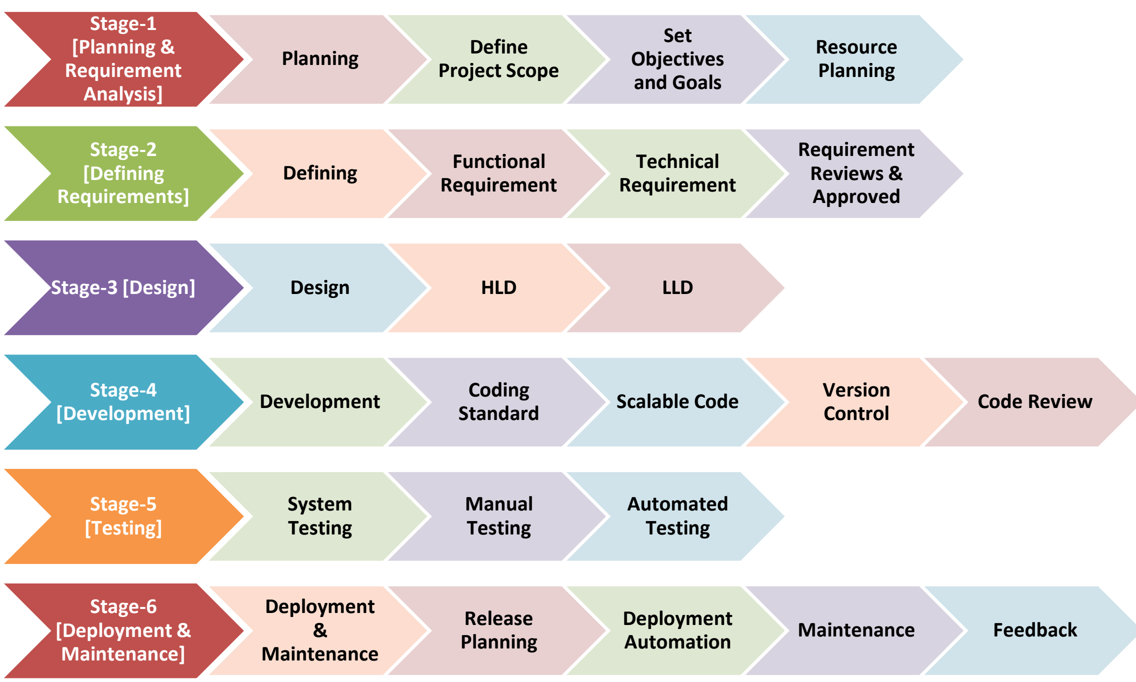
- Visibility: Stakeholders know exactly what is happening at every stage.
- Quality Control: Testing is integrated into the process, not an afterthought.
- Risk Management: Potential pitfalls are identified during the planning phase.
- Cost Estimation: Helps in predicting timelines and budgets accurately




**Stages of the Software Development Life Cycle**

SDLC specifies the tasks to be performed at various stages by a software engineer or developer. It ensures that the end product is able to meet the customer's expectations and fits within the overall budget. Hence, it's vital for a software developer to have prior knowledge of this software development process.

SDLC is a collection of these six stages, and the stages of SDLC are as follows –




**Stage 1: Planning and Requirement Analysis**

This is the most fundamental stage. Before writing code, the team must define what they are building and why

- **Activities**: Feasibility studies (technical, operational, economic), resource allocation, project scheduling, and cost estimation

- **Output**: Project Plan, Feasibility Report
- **Key Players**: Senior Engineers, Project Managers, Stakeholders


**Stage 2: Defining Requirements (SRS)**
Once the plan is approved, the specific product requirements must be defined and documented. This is done through the Software Requirement Specification (SRS) document

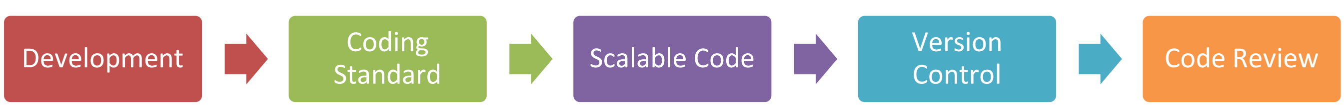
- **Activities**: Gathering detailed functional and non-functional requirements from customers
- **Output**: SRS Document (The "Bible" for the development team)
- **Key Players**: Business Analysts (BAs), Product Owners


**Stage 3: Designing Architecture**
The requirements are turned into a technical blueprint. This phase defines the overall system architecture and technology stack.

- **High-Level Design (HLD):** Defines the architecture, database design, and relationships between modules
- **Low-Level Design (LLD):** Defines the logic of individual components, API interfaces, and database tables
- **Output**: Design Document Specification (DDS)
- **Key Players**: System Architects, Lead Developers


**Stage 4: Development (Coding)**
This is the longest phase where the actual building takes place. Developers write code based on the Design Document.

- **Activities**: Coding, Code Reviews, Unit Testing, and Static Code Analysis.
- **Tools**: Compilers, Debuggers, IDEs (VS Code, IntelliJ), Version Control (Git).
- **Output**: Source Code, Executable Software.
- **Key Players**: Developers (Frontend, Backend, Full Stack)



**Stage 5: Testing**

Once the code is written, it moves to the Quality Assurance (QA) team. The goal is to find bugs before the customer does.

**Types of Testing:**

- **Unit Testing**: Testing individual functions.
- **Integration Testing**: Ensuring modules work together.
- **System Testing**: Testing the entire application flow.
- **User Acceptance Testing (UAT):** Verifying the software meets business needs.

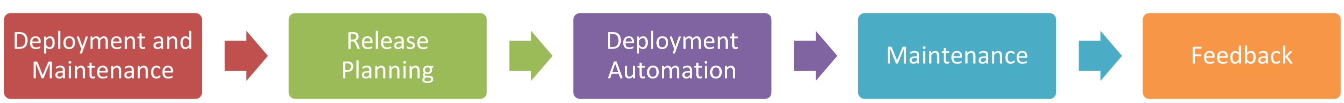
**Output**: Bug Reports, Test Cases, Quality Report.

**Key Players**: QA Engineers, Testers


**Stage 6: Deployment**

The software is released to the end-users. In modern DevOps environments, this is often automated via CI/CD pipelines

- **Activities**: Setting up production environments, deploying code, and smoke testing the live environment
- **Output**: Live Application
- **Key Players**: DevOps Engineers, Release Managers




**Stage 7: Maintenance**

The cycle doesn't end at deployment. Software must be maintained to remain useful.

- **Activities**: Bug fixing, upgrading libraries, performance tuning, and adding new features.
- **Output**: Patches, Updates, New Versions.
- **Key Players**: Support Engineers, Developers.

**Software Development Life Cycle Models**

Software Development Models are structured frameworks that guide the planning, execution, and delivery of software projects. They define the sequence of development stages, such as requirements, design, coding, testing, and deployment.

**Common Models**

- Waterfall Model
- Agile Model

- V-Model
- Spiral Model
- Incremental Model
- RAD Model

## Task to Perform

**Case Study: Online Appointment Booking System for a Hospital**

A medium-sized hospital currently manages patient appointments manually using registers and phone calls. This leads to –

- Long waiting times
- Appointment clashes
- Difficulty in tracking doctor availability

The hospital management has decided to develop on Online Appointment Booking System that allows –

- Patients to book appointments online
- Doctors to view their daily schedules
- Administrators to manage doctors and time slots

As a software engineering student, you are asked to observe the scenario and explain how the project would progress through the different phases of a software development project

**Based on the above given task students should perform the following task –**

a) **Identify the problem and project goal**
   Describe – [At least in 100 words]
   - The existing problem
   - The need for automation
   - The overall goal of the proposed software system

b) **Explain SDLC phases with respect to the case study**

   For each phase of the Software Development Life Cycle, explain what activities will be carried out for the given hospital appointment system
   - Requirement Analysis
   - System Design
   - Implementation [Coding]
   - Testing
   - Deployment
   - Maintenance

   Each phase must be explained in relation to the case study, not just in theory

c) **Deliverables and each phase**

   List at least one deliverable/output produced in each SDLC phase

**d) SDLC phase observation table**

Complete the following table –

| SDLC Phase | Key Activities [Case Specific] | Output |
|---|---|---|
| Requirement Analysis | | |
| System Design | | |
| Implementation [Coding] | | |
| Testing | | |
| Deployment | | |
| Maintenance | | |

**Submission Requirements –**

- Handwritten observation report (2-3 pages)
- Proper heading for each SDLC phase
- Neat diagrams wherever necessary
- Proper table representation

**Observations:**

- SDLC provides a disciplined approach to software development
- Each phase has specific inputs, activities and outputs
- Proper execution of early phases reduces errors in later stages
- Documentation plays a vital role throughout the SDLC

**Conclusion:**

The experiment helped in understanding the importance of SDLC and its phases in software development. A structured life cycle ensures better planning, execution and delivery of software projects with improved quality and reliability

**Expected Oral Questions:**

1) What is SDLC?
2) Why is SDLC important in software development?
3) List the phases of SDLC
4) What is the role of requirement analysis?
5) Differentiate between design and implementation phases
6) What types of testing are performed in SDLC?
7) What is meant by software maintenance?

**FAQs in Interview:**

1) What is SDLC?
2) Which SDLC phase is most efficient?
3) Is SDLC applicable to all software projects?
4) What happens if SDLC phases are skipped?
5) How does SDLC help project management?